Student ID _____ Student Name _____

University of the South Pacific

School of Information Technology, Engineering, Mathematics, and Physics

IS221 Web Applications Development

Mid Semester Test Solutions

Semester 1, 2021

Possible Marks: **100**

Weight: **20%**

Time Allowed: **110** minutes

Number of Pages: **19** (including this page)

Learning Outcomes:

2. Apply ethical practices to the development of the websites

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Instructions:

1. Sections A, B, C, D: Answer ALL questions.
2. Write ALL the answers in the spaces provided in the Answer Booklet.
3. If you require extra space for your answers, use the last page.

Student ID _____ Student Name _____

Section A: Multiple Choice 10 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question. There is only **one** possible answer.

1) Mark is given the responsibility of setting up a method for distributing information at the university where he works. The content that needs to be distributed will include course lists with student data, lecture notes, and slides that should be available only to the students in a given course. Which of the following technologies is the best choice for Mark?

   a. An internet.
   b. An extranet.
   c. An intranet.
   d. None of the above.

2) You should provide grounding for the user by placing navigation elements ____ on each page. Intranet.

   a. in the same position
   b. at the top
   c. at the left
   d. in different positions

3) A _____ is an area of the Internet a particular organization or person manages.

   a. Wireframe
   b. domain
   c. proxy
   d. protocol

4) A _____ is located at the top of a webpage and identifies the site, often by displaying the business name or logo.

   a. footer
   b. copyright label
   c. header
   d. navigation pane

5) Steve is new to web development, and he has been asked to create a website with a minimum of 10 webpages for one of his high school events using Hypertext Markup Language (HTML) 5. Steve wants to insert the copyright information in the webpages for the images used in the website. Where would you recommend him to place this information?

   a. `<header>…</header>`
   b. `<nav>…</nav>`
   c. `<main>…</main>`
   d. `<footer>…</footer>`

6) ____ paths tell the browser where a file is located relative to the document the browser currently is viewing.
   a. Absolute
   b. Relative
   c. Partial
   d. Complete

7) If one wants to format the content in the div id="container" section, then #container should be used as a(n) _____.
   a. declaration
   b. selector
   c. definition
   d. anchor

8) Mary works for an IT company that creates websites for clients. Her role entails the visual designing and creation of documents on the web for her clients. Mary could be referred to as a _____.
   a. Web designer
   b. Web author
   c. Web developer
   d. Web programmer

9) When designing navigation for mobile devices, three important considerations are:

   a. Minimize the number of clicks, regional navigation symbols and provide as many choices for the user as possible.
   b. Minimize the number of clicks, regional navigation symbols and simplify the user's choices
   c. Minimize the number of clicks, universal navigation symbols and provide as many choices as possible for the user.
   d. Minimize the number of clicks, universal navigation symbols and simplify the user's choices.

10) Mike is working on the development of a webpage for FreshGroceries. He has been asked to create a list of all the essential daily household goods that people can purchase easily through this online grocery store.

    Mike decides to use the Cascading Style Sheets (CSS) list-style properties to create the list of all the grocery goods offered by the online store. Which of the following style rules is accurate if Mike wants to define a filled square bullet for an unordered list of all the grocery items?

    a.
    ```
    uli {
    style-rule: square bullet type;
    }
    ```
    b.
    ```
    ol {
    list-style-type: square;
    }
    ```
    c.
    ```
    ul {
    list-style-type: square;
    }
    ```
    d.
    ```
    <uli/> {
    list-style-type: square;
    }
    ```

Student ID _____ Student Name _____

Section B: Matching Questions 10 Marks

Write the letter of the correct term that matches the given description in the space provided

A. <colspan>	1. ____F____works like the text input, but the entered text is hidden by asterisks.
B. World Wide Web	**2.** ____D____Monitors display colors by mixing three basic colors of light.
C. Strict	**3.** ____A____ Attribute that lets you create cells that span multiple columns of table.
D. RGB	**4.** ____G____Number of columns of Checkboxes to display
E. Marketing	**5.** ____B____ .Collectionof linked documents, graphics and sounds that can be accessed over the internet.
F. Password input box	**6.** ____E____ Responsible for promotion and attracting new customers to the website
G. RepeatColumns	**7.** ____C____ Document type that relies solely on CSS stylesheets for all layout properties
H. Wireframes	**8.** ___I_____ Shopping Cart at top right hand corner on Amazon's website
I. Designing for location	**9.** ___J____Website is perceivable, operable, understandable and robust
J. Accessibility	**10.** ___H_____ Graphic design and page template creation phase.

Student ID _____ Student Name _____

Section C: Coding Questions 30 Marks

1) The table below in Figure 1 shows the weekly time sheet for Mr. Steve an employee from the sky limit Institute (SLI).

   a) Write the appropriate HTML codes that will markup the table in Figure 1 below to make meaningful semantic table.

   [Hint: consider all those tags we've covered: the table, the row, the data, the headers, the table head, table body, table foot, caption, the col group and the col,row tags].

[10 marks]

Weekly Time Sheet

Day		Regular Hours	Overtime Hours	Sick Hours	Vacation Hours	Total
Monday	11/1/2021	8.00				8.00
Tuesday	11/2/2021	4.00		4.00		8.00
Wednesday	11/3/2021	4.00			4.00	8.00
	Total Hours	16.00	0.00	4.00	4.00	24.00
	Hourly Rate	$20.00	$30.00	$20.00	$20.00	
	Total Pay	$320.00	$0.00	$80.00	$80.00	$480.00

Figure 1: Weekly Time Sheet

ANSWER:

```
table>
    <caption id="timesheet">Weekly Time Sheet</caption>
      <colgroup>
        <col span="2" class="day">
        <col>
        <col>
        <col>
        <col>
        <col>
      </colgroup>
        <thead>
            <tr>
                <th colspan="2" scope="colgroup">Day</th>
                <th scope="col">Regular Hours</th>
                <th scope="col">Overtime Hours</th>
                <th scope="col">Sick Hours</th>
                <th scope="col">Vacation Hours</th>
```

```
            <th scope="col">Total</th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td>Monday</td>
            <td>11/1/19</td>
            <td>8.00</td>
            <td> </td>
            <td> </td>
            <td> </td>

            <td>8.00</td>
        </tr>
        <tr>
            <td>Tuesday</td>
            <td>11/2/19</td>
            <td>4.00</td>
            <td> </td>
            <td>4.00</td>
            <td> </td>

            <td>8.00</td>
        </tr>
        <tr>
            <td>Wednesday</td>
            <td>11/3/19</td>
            <td>4.00</td>
            <td> </td>
            <td> </td>
            <td>4.00</td>

            <td>8.00</td>
        </tr>
    </tbody>
    <tfoot>
        <tr>
            <td> </td>
            <th scope="row">Total Hours</th>
            <td>16.00</td>
```

```
            <td>0.00</td>
            <td>4.00</td>
            <td>4.00</td>
            <td>24.00</td>
        </tr>
        <tr>
            <td> </td>
            <th scope="row">Hourly Rate</th>
            <td>$20.00                                  </td>
            <td>$30.00                                  </td>
            <td>$20.00</td>
            <td>$20.00</td>
            <td>  </td>
        </tr>
        <tr>
            <td> </td>
            <th scope="row">Total Pay</th>
            <td>$320.00                                 </td>
            <td>$0.00</td>
            <td>$80.00</td>
            <td>$80.00                                  </td>
            <td>$480.00</td>
        </tr>
    </tfoot>
</table>
```

b) Write the appropriate style rule that renders the display as in Figure 2 below. The style rule transforms Figure 1 (in part (a)) to Figure 2 below by adding the following properties:

- All borders collapse.
- All text alignment : left.
- Table border: solid 2px black
- Table headings background color: blue
- Table heading text color: white
- caption size: 1.5rem
- caption bottom padding: 1rem
- Table cell bottom border: solid 1px black
- Table cells padding: 0.5 rem

Do not include <style type="text/css"> … </style> element. **[10 marks]**

Weekly Time Sheet

Day		Regular Hours	Overtime Hours	Sick Hours	Vacation Hours	Total
Monday	11/1/21	8.00				8.00
Tuesday	11/2/21	4.00		4.00		8.00
Wednesday	11/3/21	4.00			4.00	8.00
	Total Hours	16.00	0.00	4.00	4.00	24.00
	Hourly Rate	$20.00	$30.00	$20.00	$20.00	
	Total Pay	$320.00	$0.00	$80.00	$80.00	$480.00

Figure 2: Weekly Time Sheet- CSS

ANSWER:

```
table {
    border:2px solid black;
    border-collapse: collapse;
}
th {
    text-align: left;
    background-color: blue;
    color: white;
}
caption {
    font-size: 1.5rem;
    text-align: left;
    padding-bottom: 1rem;
}
td, th {
    padding: 0.5rem;
    border-bottom: 1px solid black;
```

2) This question is all about **ASP.NET**, especially based on Lab 5 you did in IS221. Figure 3 below shows the rendered XHTML page on the client's browser with the form being filled and the user clicked the **Register Me** button.

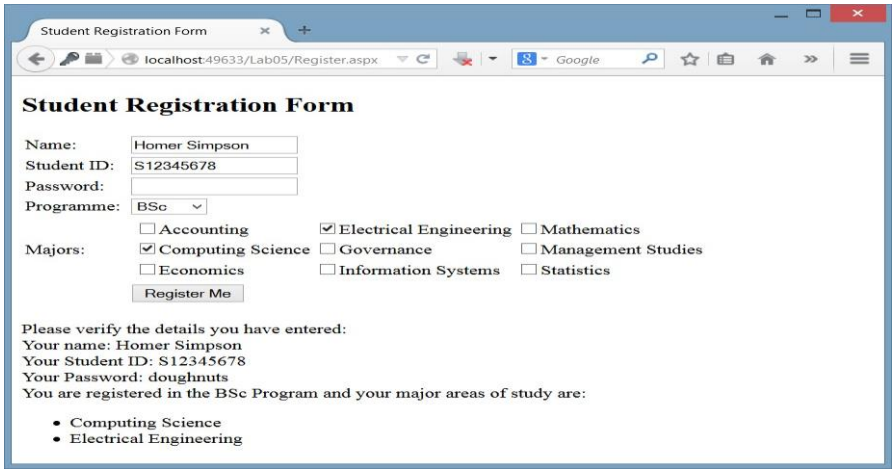


Figure 3 :Register.aspx rendered successfully on the client's browser

Note that the **Register.aspx** Web form, in addition to the text content, includes a number of ASP.NET server controls or objects as summarized in the table below:

ASP.NET control/object	Name (used in coding)
3 TextBox	**txtName, txtStudentID, txtPassword**
1 DropDownList	**ddlProgram**
1 CheckBoxList	**cblMajor**
1 Button	**btnRegister**
1 Label	**lblVerify**

Use Figure 3 together with any additional information to answer the questions that follow:

a) The **general syntax** of a ASP.NET Web server control or object is given below as well as an example using the Text Box server control. Every time you add a new server control, it comes with two important properties or attributes: **ID** and **Runat**. Other attributes can be added as required.

General Syntax:

```
<asp:TypeOfControl ID="ControlName" Runat="Server" />
```

Example:

```
<asp:TextBox ID="txtName" runat="Server"></asp:TextBox>
```

Write the ASP.NET/XHTML code for the **txtPassword** TextBox control. Note that you also need to add an additional attribute to the **txtPassword** TextBox control to mask the password as it is typed in by the user.

[3 Marks]

ANSWER:

```
<asp:TextBox ID="txtPassword" runat="server" TextMode="Password"></asp:TextBox>
```

```
1 mark for the TextMode attribute
1 mark for the ID attribute
1 mark for the rest
```

b) When the **Register.aspx** has been processed and converted to the corresponding XHTML file, the **txtName** TextBox control equivalent in XHTML is:

```
<input name="txtName" type="text" id="txtName" />
```

This is the code that will be rendered by the client's browser.
Write the corresponding XHTML code for the **txtPassword** TextBox control.

[2 Marks]

ANSWER:

```
<input name="txtPassword" type="password" id="txtPassword" />
```

```
1 mark for the type attribute.
1 mark for the rest all correct.
```

c) In the **Register.aspx** Web Form, a **DropDownList** server control or object is used. This was named as **ddlProgram**, which simply gives a list of possible programs of study a student can select from. The ASP.NET code for the **ddlProgram** DropDownList control is:

```
<asp:DropDownList ID="ddlProgram" runat="server">
    <asp:ListItem>BSc</asp:ListItem>
    <asp:ListItem>BA</asp:ListItem>
    <asp:ListItem>BCom</asp:ListItem>
    <asp:ListItem>BEng</asp:ListItem>
</asp:DropDownList>
```

Write the corresponding XHTML code for the **ddlProgram** DropDownList server control ASP.NET code given above.
[Hint: DropDownList is rendered as a **<select>** element with 2 important attributes and the individual ListItem as **<option>** element.]

[5 Marks]

ANSWER:

```
<select name="ddlProgram" id="ddlProgram">
    <option value="BSc">BSc</option>
    <option value="BA">BA</option>
    <option value="BCom">BCom</option>
    <option value="BEng">BEng</option>
</select>
```

```
1 mark for the select element.
1 mark for the name attribute.
1 mark for the id attribute.
1 mark for 2 option elements.
1  mark for 2 option elements.
```

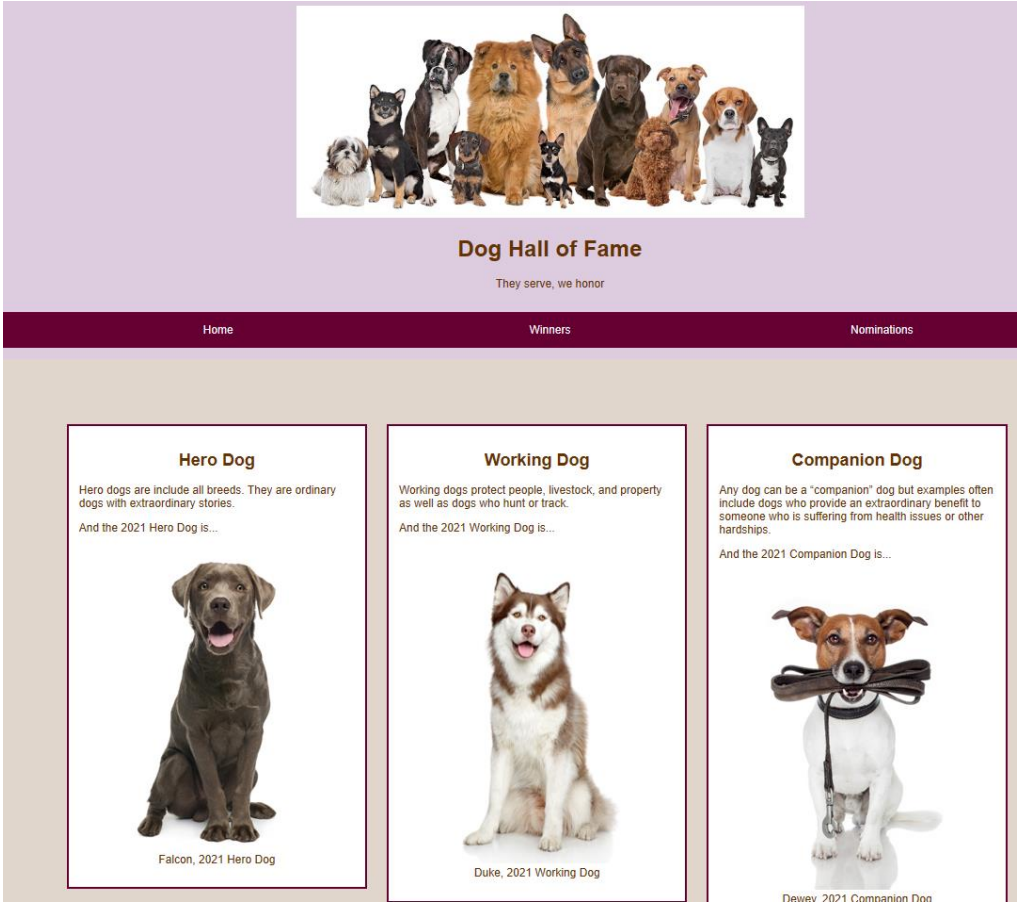
Student ID _____ Student Name _____

Section D: Short Answer Questions 50 Marks

1) Given the source code below, with the help of a diagram illustrate how the web page is going to be rendered. **[10 marks]**

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Dog Hall of Fame</title>
<meta charset="utf-8">
<style>
body {
     color: #663300;
     background-color:#ddcce0;
     font-family: Play, Cabin, sans-serif;
}
main {
     background-color: #E0D6CC;
     padding: 5%;
     display: block;
     overflow: auto;
     display: block;
     margin-top: 1em;
     clear: left;
}
section {
     margin-bottom: 2%;
     background-color: #FFFFFF;
     padding: 1%;
     border: solid 0.2em #660033;
     float: left;
     width: 28%;
     margin: 1%;
}
section h2 {
     text-align: center;
}
figure {
     text-align: center;
}
nav ul{
     padding: 0;
     margin: 0;
     list-style-type: none;
     background-color: #660033;
     color: #FFFFFF;
     text-align: center;
}
nav li {
     display: inline;
     margin-top: 1em;
     margin-bottom: 1em;
     display: inline-block;
     width: 30%;
}
nav li a{
     color: #FFFFFF;
     margin: 5%;
     text-decoration: none;
}
img {
     max-width: 100%
}
```

```
header {
     padding-bottom: 1em;
     text-align: center;
}
</style> <!-- #663300 - brown ,#ddcce0 - light purple, #E0D6CC - bage/light
brown,#FFFFFF- white,#660033 -Tyrian Purple -->

</head>
<body>
<header>
    <img src="images/banner.jpg" class="center-img" alt="banner logo for Dog Hall
of Fame" >
     <h1>Dog Hall of Fame</h1>
     <p>They serve, we honor</p>
</header>

<nav>

     <ul>
        <li><a href="dogfame08-solution.html">Home</a> </li>
        <li><a href="winners08-solution.html">Winners</a></li>
        <li><a href="nominations08-solution.html">Nominations</a></li>
     </ul>

</nav>

<main>

     <section>
     <h2>Hero Dog</h2>
     <p>Hero dogs are include all breeds.  They are ordinary dogs with
extraordinary stories.</p>
     <p>And the 2021 Hero Dog is...</p>
     <figure>
     <img src="images/heroNew.jpg" alt="Falcon hero dog">
     <figcaption>Falcon, 2021 Hero Dog</figcaption>
     </figure>
     </section>

     <section>
     <h2>Working Dog</h2>
     <p>Working dogs protect people, livestock, and property as well as dogs who
hunt or track.</p>
     <p>And the 2021 Working Dog is...</p>
     <figure>
     <img src="images/workingNew.jpg" alt="Duke working dog">
     <figcaption>Duke, 2021 Working Dog</figcaption>
     </figure>
     </section>


     <section>
     <h2>Companion Dog</h2>
     <p>Any dog can be a “companion” dog but examples often include dogs who
provide an extraordinary benefit to someone who is suffering from health issues or
other hardships.</p>
     <p>And the 2021 Companion Dog is...</p>
     <figure>
     <img src="images/compNew.jpg" alt="Dewey companion dog">
     <figcaption>Dewey, 2021 Companion Dog</figcaption>
     </figure>
     </section>

</main><!--end main-->

</body>
</html>
```


Dog Hall of Fame
They serve, we honor
Home
Winners
Nominations
Hero Dog
Hero dogs are include all breeds. They are ordinary dogs with extraordinary stories.
And the 2021 Hero Dog is...
Falcon, 2021 Hero Dog
Working Dog
Working dogs protect people, livestock, and property as well as dogs who hunt or track.
And the 2021 Working Dog is...
Duke, 2021 Working Dog
Companion Dog
Any dog can be a "companion" dog but examples often include dogs who provide an extraordinary benefit to someone who is suffering from health issues or other hardships.
And the 2021 Companion Dog is...

2) Using good coding practices ensures the greatest standards-compliance, presentation, and usefulness of the content. list with examples five rules that conforms to good coding practices. **[10 marks]**

ANSWER:

1. All tag names must be in lower case -

   <html> NOT <HTML>

2. All tags must have a corresponding matching closing tag (container tags)

   <body> </body>

3. Empty elements must have an end tag or include the closing tag

 </br> OR

4. All attribute values must be enclosed in quotes (either double or single – the notes use double)

   size="10"

5. Nested tags must be terminated in the reverse order to the way they are written.

   a. … is a correct construction,
   b. … is incorrect.

3) You have been hired to develop an Exquisite Dinig Experience's website you are required to integrate the following site requirements:
   - Logo and mission statement should be featured.
   - Search box
   - Links to the other pages of the website should be available
   - Chef's specials should be a main feature content
   - Advertisements of menu items should also feature
   - Copyright statement should feature as well
   - Links to all other pages should be available on all pages
   - Uniform look and feel for all the pages, the only difference should be the actual conten

   a) With the aid of wireframes, illustrate a potential template for the Home page.

[5 Marks]

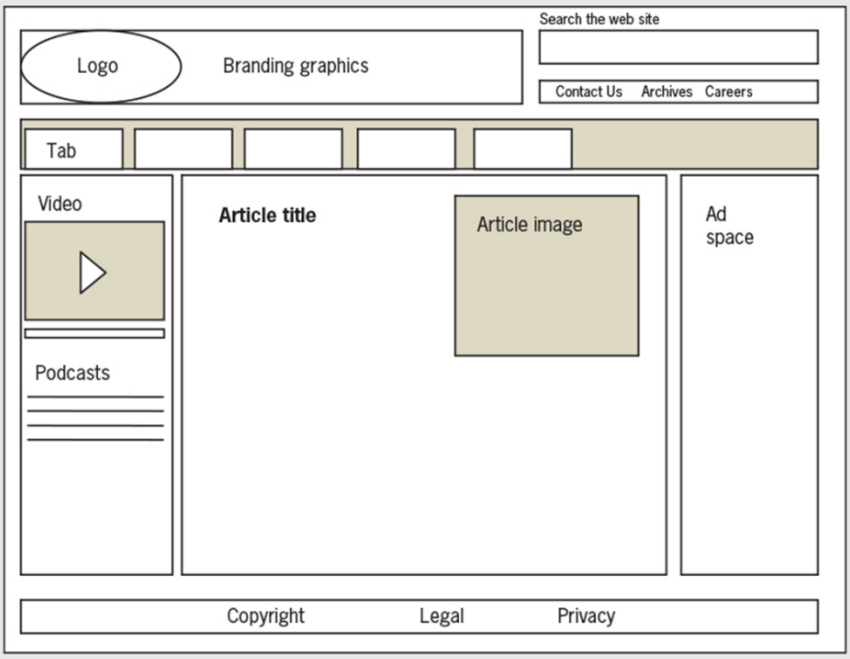
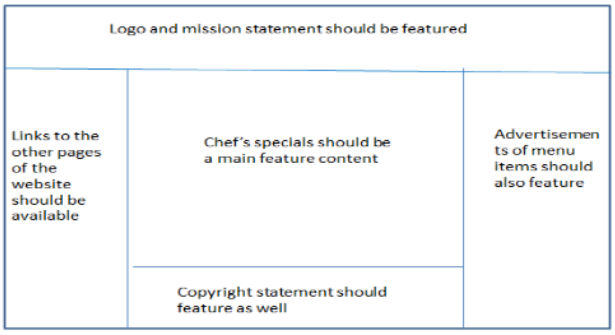
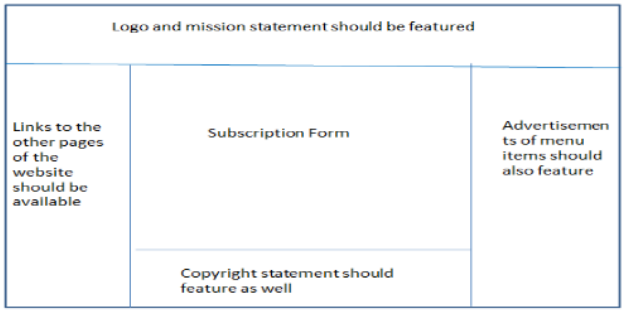


Figure 3-3: Sample wireframe for page layout
© 2015 Cengage Learning®



b) With the aid of wireframes illustrate how one of the content pages would look like.

[5 Marks]



c) How does the Exquisite Dinig Experience's website become live? **[5 Marks]**

To make your web site live, you transfer your web site files to a web server, a computer connected to the Internet and running server software. The software lets the computer use the Hypertext Transfer Protocol (HTTP) to serve HTML files to web browser clients. Unless your company or organization has a web server and hosts its own content, you must use the services of a web hosting provider. After you choose a server to host your files, you need to select file transfer software and upload the web site files from your development machine to the web server.

Student ID _____ Student Name _____

4) With reference to the Exquisite Dinig Experience's website, discuss how you would go about implementing the stages of the web development life cycle by identifying key activities and artefacts for each stage.

[15 Marks]

Stage 1_____ (1/2 Mark)
Activities (2 Marks)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Stage 2_____ (1/2 Mark)
Activities (2 Marks)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Stage 3_____ (1/2 Mark)
Activities (2 Marks)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Stage 4_____ (1/2 Mark)
Activities (2 Marks)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Student ID _____ Student Name _____

Stage 5_____ (1/2 Mark)
Activities (2 Mark2)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Requirements specification

Activities

- Determine how the customer wants to use the site e.g. will there be credit card payments etc. Will there be a catalogue for customers to view and select the kava packs they want. How about delivery, will you customers need to provide their delivery address etc.

- Type of website in this case will be an e-commerce website therefore think of common elements that an e-commerce website should have.
- Since it's going to be an e-commerce website you will need to think of an appropriate domain name e.g. renting online a .com domain would be most appropriate.
- Confirm requirements and specifications with client. Artefacts

- Registered domain name (www.uniroomsforrent.com.fj)
- Requirements specifications document which will consist of: ○ Technical specifications e.g. online searchable catalogue, form to enter customer's details etc and a database to store catalogue details and if need be customers' details.
    - *Design requirements* e.g. colour scheme or a theme to be used for the website that is in line with the Uni rooms for rent concept
    - *Page layout sketches* e.g. a sketch of what the site may look like ○ *Audience definition* e.g. who your potential customers are, will they be just local customers or international customers

Information design and taxonomy creation

` Activities

- Sorting out the web pages the customer wants for instance what content to have on what page and how to structure content and hyperlinks etc. Think in terms of the layouts discussed in readings, customer might want to have a featured product/rental specials or something

- Sketch of what the pages and the website may look like and navigation between pages etc.

Artefacts

- A sketch of the various web pages and how they should be interlinked based on their content

Graphic design and page template creation

Activities

- Creating web graphics, and page templates e.g. logo for Uni rentals should be representative of their product and what the company wants to portray etc.
- Navigation buttons/scheme etc.
- Wireframe sketches of web site/web pages Artifacts5
- Web graphics, web page templates

Construction and Web development/Quality Assurance and User Testing

Activities

- Coding of the required elements and content
- Construction of databases
- Testing of coded elements to ensure they are functioning correctly e.g. testing the room booking functionality etc.

Artefacts

- A database driven functioning website with forms, catalogues and components that are tested and functional.

Publishing and promotion

Activities

- Determining whether to host the website onsite (your own servers etc) or use a web hosting company.
- Web site is uploaded to web server and is now available and live
- Use Facebook for business to promote website or twitter etc

Artefacts

- Fully functional website

Maintenance

Activities

- Updating the website e.g. adding new products, on sale items etc.

Artefact

- Well maintained site that is available, up to date and fully functional